

Analytical Study of Ankita Cashew Processing

A Proposal report for the BDM Capestone Project

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Declaration Statement

I am working on a Project titled “Analytical Study of Ankita Cashew Processing”. I extend my appreciation to **Ankita Cashew Processing**, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through primary sources and carefully analyzed to assure its reliability.

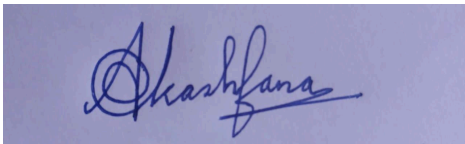
Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the information of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all the recommendations are business-specific and limited to this project exclusively, and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.

Signature of Candidate:



Name: Akash Jana

Date: February 11, 2026

1. Executive Summary

Ankita Cashew Processing is a cashew processing company in West Bengal, India. It operates in B2B segment and supplies processed cashew kernels to vendors across country. Its sister concern, Adrita Cashew Processing, operates in the same domain. Together, both companies are valued at 10 crore INR, with daily production capacity of around 5 metric-tons.

The key business problems include unpredictable RCN prices, fluctuating market demand, disorganised inventory, processing waste, delayed buyer payments, and the absence of real-time profit-and-loss tracking. Manual operations create a risk of hidden losses.

To address these problems, the project uses a structured analytical approach. Sales forecasting through time-series and regression models supports demand and price planning. Batch-wise yield analysis will reduce wastage and improve inventory control. A cost modelling framework will provide real-time financial visibility. Buyer payment analysis and risk scoring will help manage credit risk. Anomaly detection will identify and prevent hidden losses.

The tools used are Microsoft Excel for data organization and pivot analysis, Python via Jupyter Notebook for regression, ARIMA forecasting, logistic regression, and anomaly detection, scikit-learn, and statsmodels, and a Flask dashboard for live profit-loss monitoring.

The expected outcome is better inventory management, reduced wastage, improved cash flow, and more stable profitability.

2. Organization Background

Ankita Cashew Processing is a cashew processing business based in West Bengal, India, owned and managed by Usha Jana. The company buys raw cashew nuts, processes them into graded kernels, and supplies them to vendors across the country. It operates primarily in the B2B segment and focuses on quality, consistency, and timely delivery.

Its sister concern, Adrita Cashew Processing, is owned and managed by Mithu Jana and operates in the same line of business. Together, both companies are valued at approximately 10 crore INR and maintain a combined daily production capacity of around 5 metric tons.

Both businesses have grown steadily in scale and market presence. Their operations are largely experience-based, and they are now moving towards structured systems and data-driven methods to improve efficiency and overall profitability.

3. Problem Statements

3.1 Demand and Price Volatility

Problem Statement: To manage demand and raw cashew price fluctuations effectively to improve purchasing decisions, stabilise selling prices, and protect profit margins.

Company faces frequent changes in market demand and RCN prices. These fluctuations leads difficult decisions when to buy raw materials and at what price to sell finished products. Sometimes the company purchases stock at high prices and later sells at lower because the market drops. These uncertain profit margins make financial planning challenging.

3.2 Inventory and Wastage Loss

Problem Statement: To reduce inventory waste and processing losses by implementing systematic batch tracking and yield analysis to improve operational efficiency.

The business faces issues with poorly managed inventory. Without proper tracking, raw cashews and processed kernels may remain in storage for too long, leading to quality loss. During processing, wastage may also occur due to breakage, moisture loss, or handling errors. These small losses, when accumulated over time, can significantly reduce overall profitability.

3.3 Real-Time Profit and Loss Tracking

Problem Statement: To develop a real-time profit-loss tracking system providing clear financial visibility across batches, grades, and cost heads.

Currently, there is no system to track profit-loss in real time. Costs such as raw material, labour, electricity, and transportation are not always calculated per batch or per kilogram. As a result, the company may not clearly know which grades or orders are more profitable. This lack of financial visibility leads to weak and uninformed decision-making.

3.4 Buyer Risk and Payment Delays

Problem Statement: To assess buyer payment risk and reduce credit exposure by systematically analyzing payment history and developing a buyer risk scoring mechanism.

The Company operates in a B2B model, and it depends on buyers for timely payments. Some buyers may delay payments, which affects cash flow. Without a proper system to track payment history and buyer reliability, the business may continue extending credit to high-risk customers, increasing financial pressure on the organization.

3.5 Hidden Losses and Fraud Prevention

Problem Statement: To detect and prevent hidden financial losses arising from manual handling errors, weight discrepancies, and inventory mismatches through system checks and anomaly detection.

In manual systems, hidden losses such as weight manipulation, inventory mismatches, or procurement leakages occur and may go undetected for extended periods. Such irregularities can reduce profits and create inefficiencies. A systematic approach to data checks and anomaly detection is required to identify and address such losses early.

4 Background of the Problem

4.1 Demand and Price Volatility

Raw cashew prices change based on season, weather, and export demand, and processed kernel demand also varies across the year. Since purchasing decisions are experience-based rather than data-driven, the company risks buying at poor prices or stocking grades that move slowly, directly affecting profit margins and working capital.

4.2 Inventory and Wastage Loss

There is no batch-wise inventory tracking system in place. This makes it difficult to monitor stock age, measure yield rates per cycle, or detect batches with abnormally high wastage. Since cashew kernels are perishable and sensitive to moisture, prolonged storage without monitoring raises the risk of quality loss.

4.3 Real-Time Profit and Loss Tracking

Financial records capture total costs across raw material, labour, electricity, packaging, and transportation, but these are not linked to individual batches or grades. As a result, it is not possible to identify which grades or batches are profitable and which are generating losses, limiting the scope for corrective action.

4.4 Buyer Risk and Payment Delays

The company supplies multiple vendors on credit. Some buyers delay payments beyond agreed timelines, and without a structured credit monitoring system, there is no way to distinguish reliable buyers from risky ones. This weakens working capital management and overall financial stability.

4.5 Hidden Losses and Fraud Prevention

Manual procurement and inventory processes are vulnerable to errors and manipulation. Weight discrepancies, inaccurate batch recording, and small daily leakages can accumulate into significant losses over time and often go undetected without automated reconciliation.

5. Problem Solving Approach

5.1 Data Collection

Primary data was collected directly from the operations of Ankita Cashew Processing over approximately 31 days, covering March 11 to April 11, 2026. The data spans five categories: sales records including grade-wise volumes, selling prices, and buyer details; procurement records including purchase dates, quantities, prices, and supplier information; inventory and batch records including input quantities, grade-wise outputs, processing dates, and storage durations; cost records covering labour, electricity, packaging, and transportation per batch or period; and buyer payment records covering invoice dates, amounts received, outstanding balances, and buyer-wise transaction history. Data was gathered through direct interaction with the proprietor and through physical and digital business records maintained by the company. Retrospective historical records were also collected wherever available to support trend and pattern analysis.

5.2 Methods

The following analytical methods and techniques will be employed to address each identified problem:

Demand and Price Forecasting

Time-series decomposition will be used to identify trend and seasonal patterns in sales and price data. ARIMA modelling will support short-term forecasting, with linear regression serving as a baseline model. Forecast accuracy will be evaluated using RMSE and MAE.

Inventory and Yield Analysis

Batch-wise yield will be calculated as $(\text{Grade Output} / \text{Raw Input}) \times 100$. ABC classification will prioritise high-value grades for closer monitoring. Variance analysis will compare expected and actual yield per batch, and Z-score based outlier detection will flag batches with abnormal yield rates.

Profit and Loss Modelling

Per-kilogram and per-batch cost sheets will be constructed for each grade. Contribution margin and break-even analysis will assess financial viability at different output levels. Sensitivity analysis will measure how price changes affect profit margins, and a dashboard will provide real-time P&L visibility.

Buyer Risk and Credit Analysis

Payment ageing analysis will categorise outstanding amounts into 30, 60, and 90-day buckets. Buyers will be scored based on payment delay history and order volume. Logistic regression will estimate the probability of future payment delays, and credit limits will be recommended based on individual risk scores.

Anomaly Detection for Hidden Losses

Procurement, inventory, and sales records will be reconciled against each other. Z-score thresholding and the Isolation Forest algorithm will detect irregular patterns, with alerts generated for suspicious or inconsistent transactions.

5.3 Analysis Tools

The following tools will be used across the analytical tasks described above:

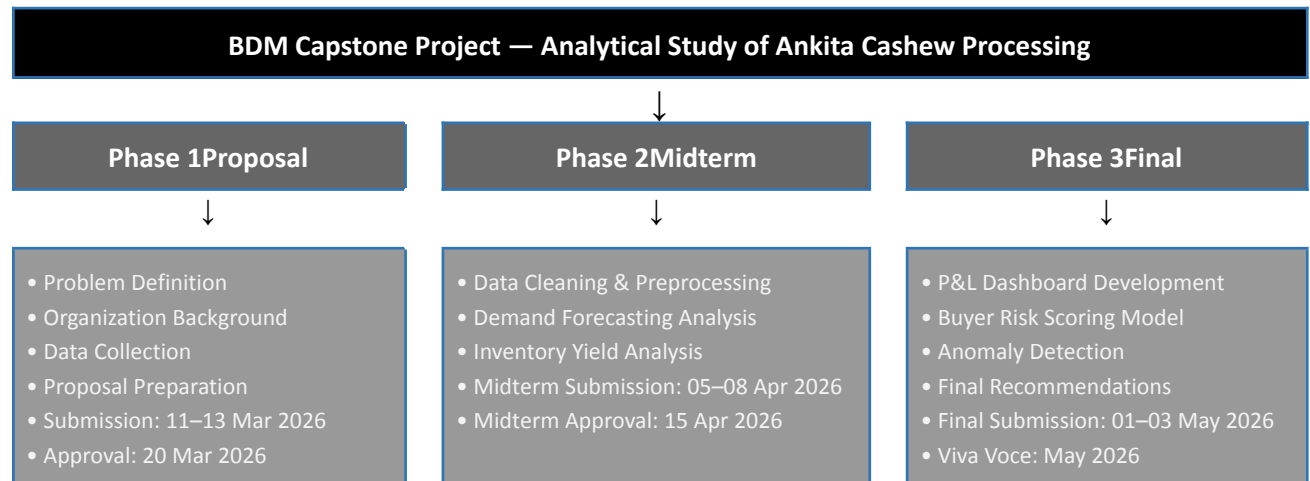
Tool	Purpose and Application
Microsoft Excel	Data organisation, cost sheets, pivot analysis, ABC classification, ageing analysis, break-even and sensitivity modelling, and chart generation.
Python (Jupyter / Kaggle)	ARIMA and linear regression for forecasting, logistic regression for buyer risk, Z-score and Isolation Forest for anomaly detection, and yield variance analysis using pandas, numpy, scikit-learn, statsmodels, and matplotlib.
Flask (Python)	Development of a basic real-time P&L dashboard for monitoring batch-wise and grade-wise margins and operational costs.

6. Expected Timeline

6.1 Work Breakdown Structure

The following flowchart presents the work breakdown structure of the BDM Capstone Project, organized across three phases: Proposal, Midterm, and Final Submission.

Each phase includes specific deliverables related to data collection, analysis, and reporting. The approval stage follows each submission before the next phase begins.



6.2 Gantt Chart

The Gantt chart below presents the scheduled timeline for all project activities, including submission dates, approval milestones, and the viva voce examination. All durations are expressed in weeks across January 2026 to May 2026.

Task	Jan 2026	Feb 2026		Mar 2026			Apr 2026			May 2026	
	W3-4	W1-2	W3-4	W1	W2	W3-4	W1	W2-3	W4	W1	W2+
Key Dates	11 Jan	01 Feb	15 Feb	01 Mar	09 Mar	16 Mar	01 Apr	09 Apr	22 Apr	01 May	08 May
Data Collection (11 Jan – 15 Feb)											
Data Cleaning (25 Jan – 11 Mar)											
Proposal Preparation (15 Feb – 11 Mar)											
Proposal Submission — 11–13 Mar 2026											
Proposal Approval — 20 Mar 2026											
Preliminary Analysis (13 Mar – 05 Apr)											
Business Analysis (20 Mar – 01 May)											
Midterm Submission — 05–08 Apr 2026											
Midterm Approval — 15 Apr 2026											
Final Report (15 Apr – 01 May)											
Final Submission — 01–03 May 2026											
Final Approval & Viva Voce — May 2026											

Task Duration

Submission

Approval

7. Expected Outcome

7.1 Improved Demand Forecasting and Sales Planning

The business will develop a clearer understanding of which cashew grades sell more frequently and how demand shifts across different months. This will support better production planning, more informed raw material purchasing, and reduced risk of overstocking or understocking.

7.2 Better Inventory Management and Reduced Wastage

Batch-wise yield analysis and stock tracking will help identify areas of excess inventory and processing loss. This will improve storage planning, reduce unnecessary wastage, and raise overall operational efficiency.

7.3 Clear Profit and Loss Visibility

A structured cost model and real-time P&L dashboard will provide batch-wise and grade-wise profit clarity. Management will be able to identify high-margin and low-margin segments and make more informed pricing and operational decisions.

7.4 Improved Cash Flow and Buyer Risk Control

will identify customers with poor payment patterns, strengthening credit control policies and reducing financial exposure from high-risk buyers.

7.5 Long-Term Data-Driven Growth Strategy

This project will build a foundation for future automation and business intelligence within the organisation. The models and dashboards developed here can be scaled as the business grows, supporting sustainable and data-informed expansion.